UNIVERSITY OF BOTSWANA

CSI473 Project Design Report

TRANS-UB - University of Botswana Student Marketplace Hub

Team 9 | Phase 1 Project Design Report

Field	Complete
Assignment/phase	Phase 1 Project Design Report
Approved project title	TRANS-UB - University of Botswana Student Marketplace Hub
Approval decision/date	Approved with minor conditions; decision date recorded in approval artefact
Team number	9
Members and IDs	Letlhabile Xoliso Serufo - 202302072 Lebogang Rantlheng - 202400279 Tshegofatso Botho Wakwena - 202302811 Amanda Mapete - 202303699 Lame Vambu - 202203427
Repository URL	https://github.com/lebogang00279/Trans-UB
Submission tag	To be added at submission
Date	11 September 2026

1. Executive summary

TRANS-UB is a student-to-student marketplace proposed for the University of Botswana. The project addresses the fragmented way students currently advertise goods through WhatsApp groups, Facebook posts, notice boards and word of mouth. The design centralises approved listings, supports search and comparison, records reservation/order status, and builds seller reputation from completed transactions.

The approved first implementation is restricted to low-risk physical goods. Food, medicines, alcohol, stolen property, restricted products and high-risk services are outside the prototype scope. Student-ID verification is simulated using synthetic data and must not be presented as integration with official University of Botswana records.

The main project decision is D-001: TRANS-UB does not process online payments and does not control the physical handover. The platform records the order and its state, while the buyer and seller arrange payment and handover directly. This reduces technical, regulatory and security risk while preserving accountability through the order lifecycle.

Phase 1 is centred on UC-05, Reserve an available listing. The design checks buyer eligibility and listing availability, rechecks availability at confirmation, creates a Pending order, changes the listing to Reserved, records the status change and notifies the seller. The same domain responsibilities are reused for acceptance, rejection, expiry, cancellation, delivery, completion and review eligibility.

2. Problem evidence, stakeholders and scope

2.1 Problem statement and evidence

The repository evidence log records screenshots from an active UB student buy-and-sell WhatsApp group, interviews with five student sellers and five student buyers, and Facebook posts advertising items to UB students. Together they show that student trading is active but fragmented, seller trust is difficult to judge, and similar offers cannot easily be compared side by side.

Evidence	Origin	What it shows
WhatsApp buy-and-sell group screenshots	7 Aug 2026; student-run group	Active student trading but little structure, no persistent seller profile and listings that disappear as chat moves.
Interviews with 5 sellers and 5 buyers	7 Aug 2026; UB main campus	Sellers reported visibility/trust problems; buyers reported unreliable sellers and uncertainty about safe sellers.
Facebook sales posts	7 Aug 2026; public group sample	Similar goods are promoted in separate posts at different prices, making comparison difficult.

2.2 Stakeholders

Stakeholder	Interest / responsibility
Student Buyer	Find and compare trustworthy listings; reserve an item; confirm receipt; review completed transactions.
Student Seller	Create/manage listings; respond to orders; arrange offline handover; build visible reputation.
Platform Administrator	Approve/remove listings, handle reports and suspend accounts where required.
SRC	Assist with escalation of serious unresolved disputes.
University Management	Policy and liability interest; no direct operational access in the current system context.

2.3 Scope and approval conditions

- Only approved low-risk physical goods are included in the first implementation.
- Food, medicines, alcohol, stolen property, restricted products and high-risk services are excluded.
- Payments and physical handovers occur outside the platform.
- Public campus handover locations should be recommended.
- Student-ID verification is simulated using synthetic data.
- Reservation expiry, rejection, cancellation, seller no-response, reporting and dispute handling are explicit workflow concerns.

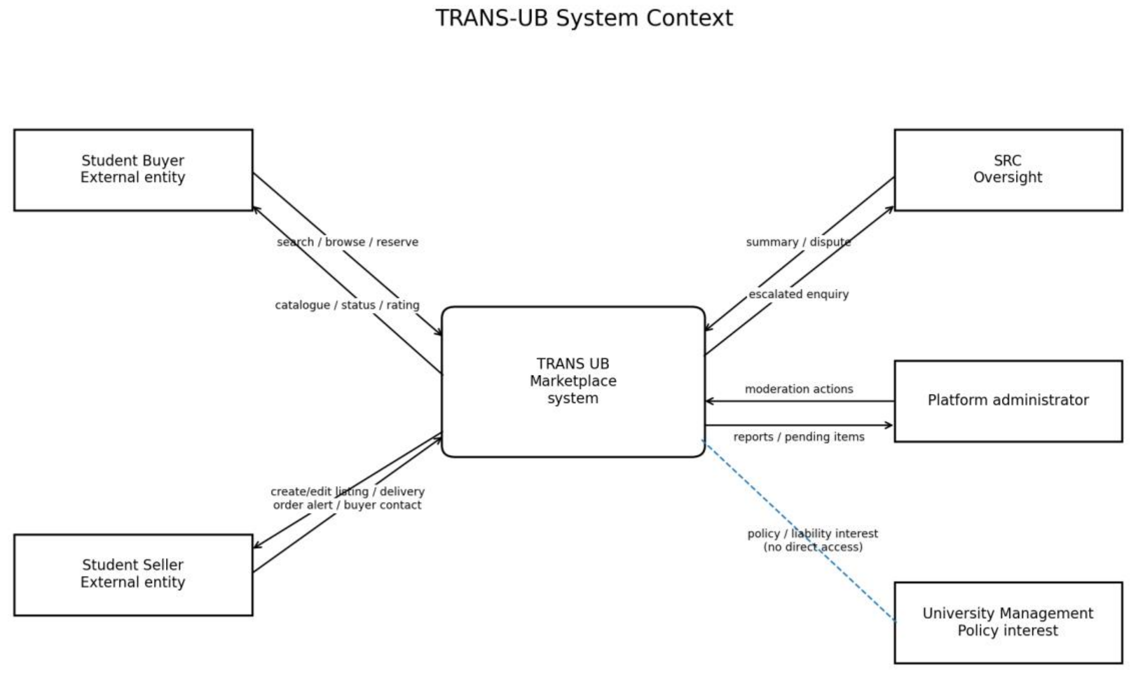

Figure 1. System context re-rendered from the repository source Models/system-context.mmd.

3. Glossary

Term	Definition
StudentAccount	Prototype account representing an eligible student buyer or seller. Administrative moderation is handled by the separate PlatformAdministrator role.
Listing	Approved low-risk item offered by a student seller.
Available	Listing state in which the item may be reserved.
Reserved	Listing state in which a live transaction prevents a second active reservation.
Order	Transaction record connecting one buyer, one seller and one listing.
Pending	Order has been created and is waiting for seller response.
Accepted	Seller has accepted the order and the parties may proceed to offline handover.
Rejected	Seller declined the order; listing is released.
Expired	Seller did not respond within the response deadline; listing is released.
Cancelled	Buyer cancelled before handover; listing is released.
Delivered	Seller records that physical handover took place.
Completed	Buyer confirms receipt; transaction becomes eligible for a review.
Review	Single rating/comment linked to a Completed order.
Notification	Recorded notice generated by the order workflow.
OrderStatusChange	History entry recording movement from one order state to another.
Synthetic verification	Prototype verification using fabricated test data rather than official UB systems.
PlatformAdministrator	Analysis role responsible for listing moderation, report resolution, account suspension where permitted, and escalation of serious unresolved disputes.
Report	A user-raised issue linked to a listing, user or order. Its lifecycle is Open -> Under Review -> Resolved or Escalated.
Sold	Terminal listing state reached after the associated order becomes Completed.

4. Functional requirements and use cases

ID	Functional requirement
FR-01	Register and log in using prototype accounts with simulated Student-ID verification.

ID	Functional requirement
FR-02	Seller creates/edits a listing for an approved low-risk physical good.
FR-03	Administrator approves, rejects or removes listings.
FR-04	Buyer searches, browses, filters and compares approved listings.
FR-05	Buyer reserves/orders an Available listing; successful confirmation changes it to Reserved and creates one Pending order.
FR-06	Seller accepts or rejects a Pending order; rejection releases the listing.
FR-07	Unanswered orders expire after 24 hours and buyers may cancel before handover; the listing is released.
FR-08	Seller marks Delivered and buyer confirms receipt to reach Completed.
FR-09	Buyer may submit one review only for a Completed order.
FR-10	Users report a listing/user/order; administrator resolves or escalates serious unresolved disputes.

4.1 Use-case model

The final Phase 1 use-case model is restricted to the approved FR-01 to FR-10 baseline. It shows Student Buyer, Student Seller, Platform Administrator and SRC only where a requirement assigns a responsibility. Unsupported oversight functions from earlier drafts have been removed so the model now matches the requirements baseline.

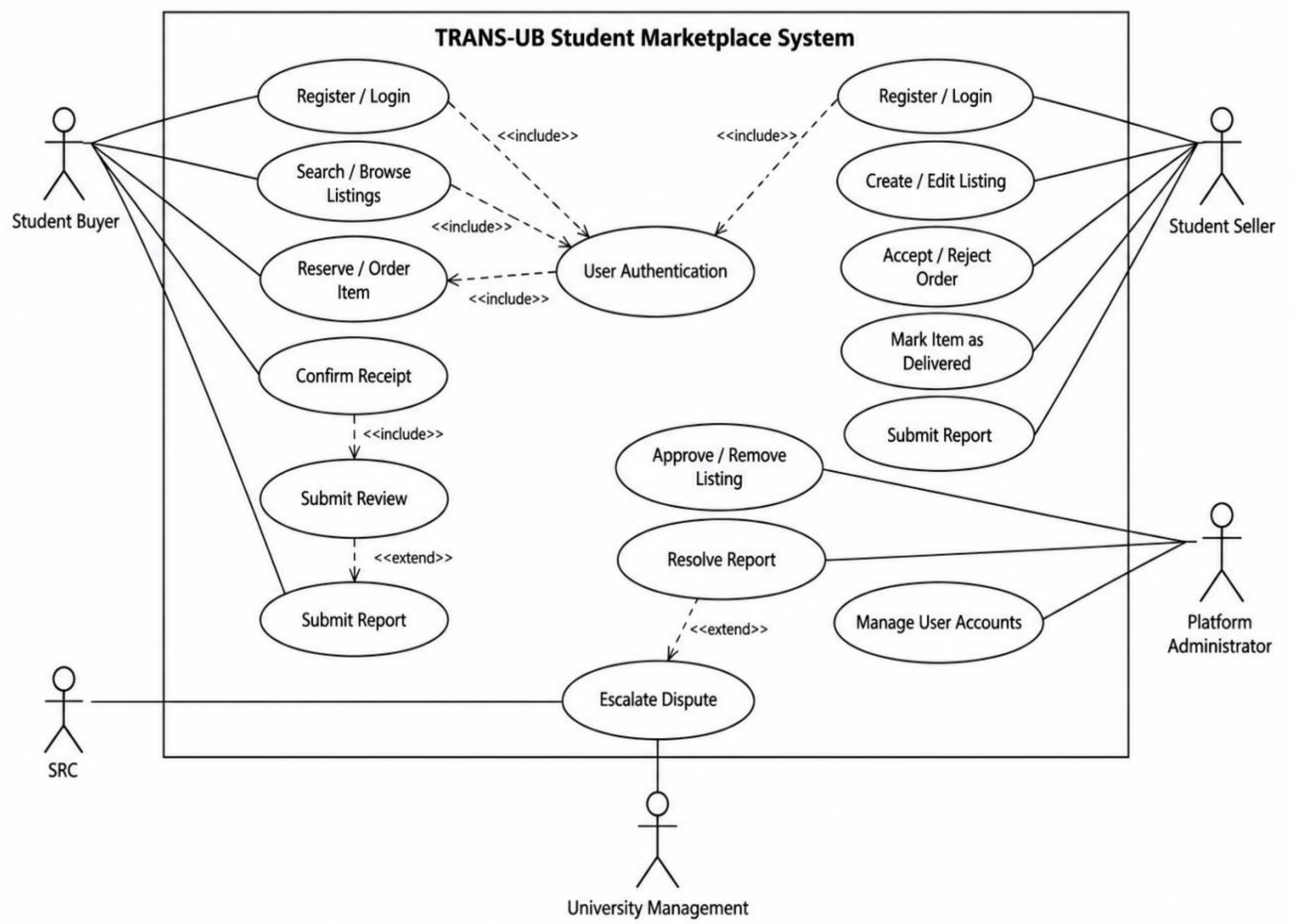

Figure 2. Corrected Phase 1 use-case baseline to be synchronized with Models/usecasemodel.drawio and Models/usecasemodel.svg.

4.2 Core use case: UC-05 Reserve an available listing

Item	UC-05 summary
Primary actor	Student Buyer
Trigger	Buyer asks to reserve/order an approved Available listing.

Item	UC-05 summary
Preconditions	Buyer is eligible and active; listing is approved and Available; buyer is not the seller.
Main success outcome	Exactly one Pending Order is created, Listing changes Available -> Reserved, response deadline is recorded and seller is notified.
Important alternative	Listing is no longer Available when buyer confirms; no Order is created and the current Listing state is preserved.
Failure-handling principle	A failed seller notification does not roll back a successful reservation.

4.3 Acceptance criteria

- An eligible buyer can reserve an approved Available listing and receives an order reference, status and response deadline.
- The system rechecks listing availability when the buyer confirms because the earlier view may already be stale.
- Only one live reservation may exist for the same listing at a time.
- If the listing is no longer Available, no new Order is created.
- Seller rejection, 24-hour expiry or buyer cancellation releases the listing.
- A review is refused until the associated order is Completed.

5. Quality requirements

Quality concern	Expected behaviour	Design obligation
Search responsiveness	Relevant results should be displayed quickly enough for interactive use.	Keep listing retrieval/filtering simple and design searchable fields deliberately.
Access control	Protected listing/order actions are denied to ineligible or suspended accounts.	Eligibility check is explicit in the reservation sequence.
Order reliability	Valid reservation confirmation must not create half-complete state.	Create Order and reserve Listing as one consistent unit.
Duplicate prevention	Two buyers confirming simultaneously must not create two live reservations.	Recheck current listing state at confirmation; only one Available -> Reserved transition may succeed.
Review integrity	Reviews are accepted only for Completed transactions.	Review is 0..1 per Order and Completed is the eligibility guard.
Notification tolerance	A delivery failure must not invalidate a successfully stored reservation.	Notification records attempt/outcome; Order remains Pending.

6. Analysis/domain model and responsibilities

The repository domain model was created by filtering candidate concepts against the project's scenarios and rules. Solution-domain concepts such as screens, pages, storage mechanisms and payment were deliberately excluded. Decision D-001 also removes Payment and Handover as domain classes because those activities take place outside the platform.

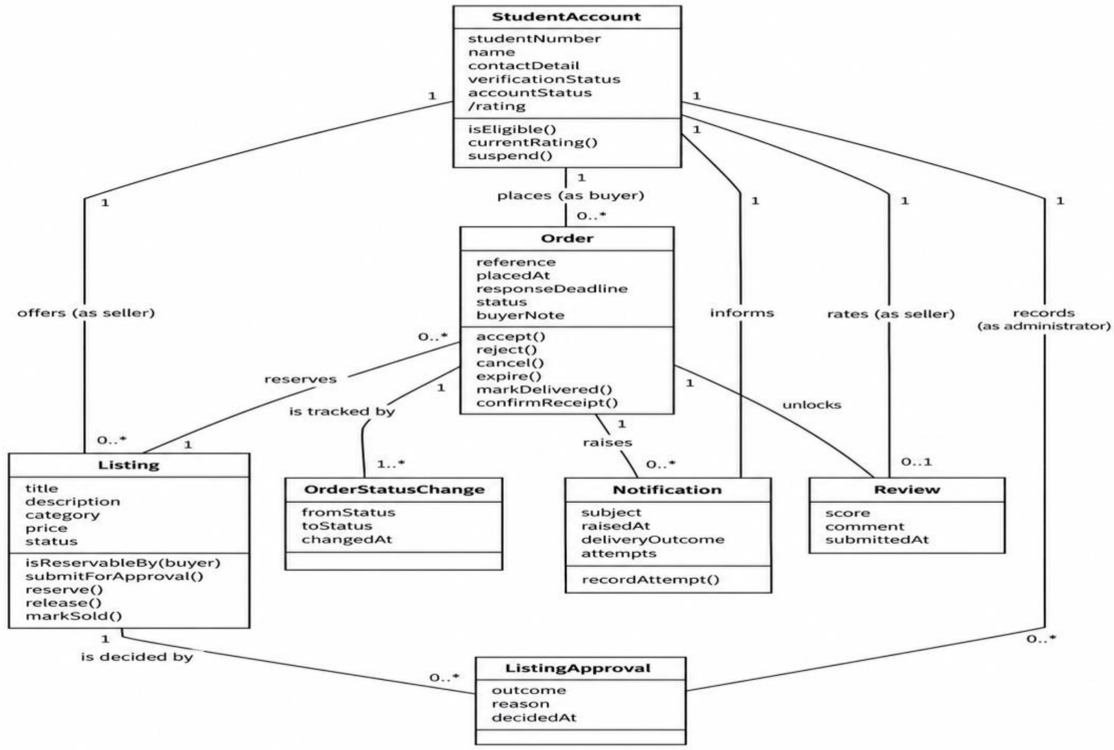

Figure 3. Corrected Phase 1 domain baseline to be synchronized with Models/domain-model.mmd and Models/domain-model.svg.

Element	Responsibility
StudentAccount	Stores identity/verification/account state; answers isEligible(); can be seller or buyer.
Listing	Owns listing status and reservation rules; answers isReservableBy(buyer), reserve(), release() and markSold().
ListingApproval	Records an administrator's approval decision and reason.
Order	Owns the transaction lifecycle and response deadline.
OrderStatusChange	Provides status history and accountability.
Notification	Records notices, delivery outcome and retry attempts.
Review	Captures one completed-transaction rating/comment and contributes to seller rating.
ReserveListingControl	Coordinates interactions; it does not own the business rules.
PlatformAdministrator	Approves/removes listings, resolves reports, records moderation decisions and escalates serious unresolved disputes.
Report	Stores report target, reason, status and resolution; supports Open -> Under Review -> Resolved/Escalated.

6.1 Responsibility allocation rationale

The repository explicitly chooses to keep rules on the entities rather than placing policy and workflow logic inside one ReservationManager. Listing decides whether it is reservable because it knows its approval/status and seller. StudentAccount decides eligibility. Order owns lifecycle transitions. ReserveListingControl therefore coordinates the use case instead of becoming a large object that reads and writes every other object.

A realistic alternative is a single ReservationManager that checks account status, checks the listing, writes the order/history and sends the notification. It would be quicker initially, but it would centralise policy, workflow and delivery concerns and grow as FR-06, FR-07 and FR-08 add more transitions. The project accepts a more message-rich design for clearer responsibilities and reuse.

7. Interaction and lifecycle

UC-05 is the main Phase 1 interaction because it creates the transaction that later use cases act on. The sequence deliberately checks the listing twice: once before showing reservation terms and again when the buyer confirms. The second check addresses stale pages and simultaneous buyers.

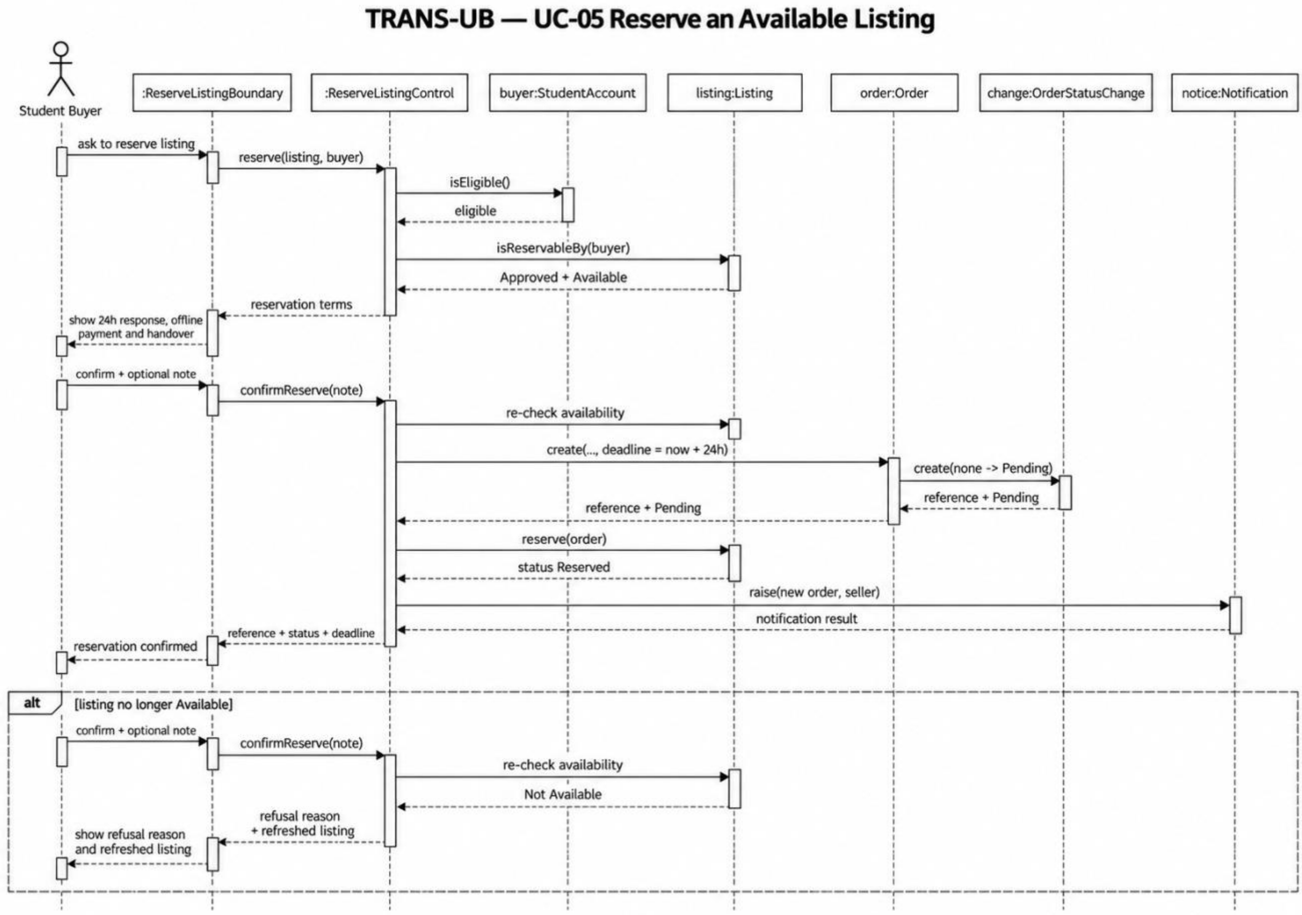

Figure 4. Core reservation sequence re-rendered from the repository source Models/sequence-core-use-case.mmd.

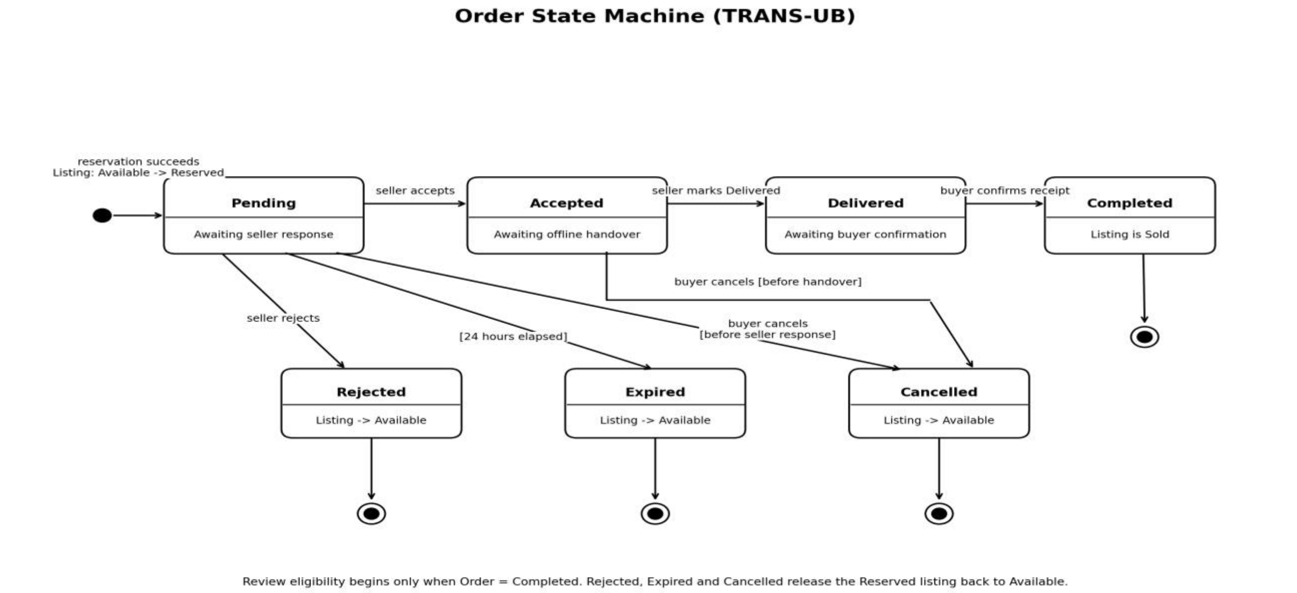

Figure 5. Corrected Phase 1 order state machine to be committed as Models/order-state-machine.mmd and Models/order-state-machine.svg.

7.1 Consistency across artefacts

Requirement	Use-case event	Analysis responsibility	State effect	Verification
FR-05	UC-05 confirmation	confirmReserve(note); Order.create(...); Listing.reserve(order)	Order -> Pending; Listing Available -> Reserved	Create one order and block a second live reservation.
FR-06	Seller response	Order.accept() / Order.reject()	Pending -> Accepted or Rejected	Rejected path releases the Listing.

Requirement	Use-case event	Analysis responsibility	State effect	Verification
FR-07	No response / cancellation	Order.expire() / Order.cancel()	Pending -> Expired/Cancelled	Listing returns to Available.
FR-08	Handover confirmation	Order.markDelivered(); Order.confirmReceipt()	Accepted -> Delivered -> Completed	Completed transaction unlocks review.
FR-09	Review	Review creation after completion	Order Completed; Review 0..1	Reject review before completion or second review.
FR-10	Submit / resolve report	Report.open(); PlatformAdministrator.resolveReport() / escalate()	Report Open -> Under Review -> Resolved/Escalated	Create report, record admin action, and verify final resolution/escalation.

8. Architecture and decisions

8.1 Decision D-001 - Offline payment and physical handover

Decision: TRANS-UB does not process online payments. Payment and physical exchange of goods are arranged directly between buyer and seller outside the platform. The system records the order, status changes and confirmation of receipt, but it does not handle, store, verify or transfer money.

Rationale: payment integration would add technical complexity, security obligations, regulatory considerations and potential financial liability that are not appropriate for a semester project. Students already use cash and mobile money, so the design supports existing behaviour rather than replacing it.

Alternative	Decision	Reason
Full online payment integration (Stripe/PayPal)	Rejected	Adds external integration, financial security/regulatory responsibilities and liability.
Mobile-money integration	Rejected	Provider-specific integrations plus payment/fraud handling increase complexity.
Escrow	Rejected	Requires TRANS-UB to hold/release money and assume additional legal and financial responsibilities.
No online payments	Selected	Keeps focus on discovery, reservation, transaction recording and reputation.

8.2 Consequences of D-001

- TRANS-UB cannot independently prove that money changed hands.
- Buyer and seller honesty/cooperation remain necessary for the offline part of the transaction.
- Accountability is created through seller Delivered and buyer Completed confirmation.
- Only Completed transactions are eligible for reviews.
- Order history supports later report/dispute handling.
- User-facing wording must clearly state that TRANS-UB does not guarantee payment or physical handover safety.

8.3 Architecture drivers for Phase 2

The Phase 2 architecture must be driven by project quality scenarios and constraints rather than by a preferred framework name. The three highest-impact drivers are: (1) transaction consistency and duplicate-reservation prevention, because only one live reservation may exist for a listing; (2) access control and trust integrity, because protected actions require eligible accounts and reviews require Completed orders; and (3) reliability under partial failure, because stale listing state or notification failure must not corrupt a valid reservation.

Two realistic architecture alternatives will be carried into Lab 7. Alternative A is a simple layered monolith with UI, application, domain and persistence layers in one deployable application. Alternative B is a modular layered monolith that keeps the same single deployment but separates marketplace, order, moderation and notification modules behind explicit interfaces. Alternative B adds structural discipline and clearer ownership at the cost of more internal interfaces; it is the stronger current candidate because it supports the responsibility boundaries already present in the analysis model without introducing distributed-system complexity.

9. Data, API/service and deployment design

The repository is still primarily at the analysis/design stage. The domain model identifies the logical records the eventual data model must support: StudentAccount, Listing, ListingApproval, Order, OrderStatusChange, Notification and Review. FR-10 is represented by a Report record with an explicit lifecycle and PlatformAdministrator responsibility. The final logical schema and service/API contracts should preserve these multiplicities and lifecycle constraints.

Design constraint	Required behaviour
Order + Listing reservation	Must be changed consistently so a stored Pending order cannot exist with Listing still Available.
OrderStatusChange	Every Order must have at least its creation entry; later transitions append history.
Review uniqueness	At most one Review per Order; Order must be Completed.
Notification failure	Outcome is separate from Order validity so notice can be retried without rolling back reservation.
Synthetic identity	No dependency on official UB records should appear in deployment/API design.

10. Detailed design, patterns and framework

The most visible Phase 1 detailed-design choice is boundary-control-entity separation in the reservation sequence. ReserveListingBoundary handles actor-facing interaction, ReserveListingControl orders the workflow, and the domain entities own their rules. This is a responsibility-centred design rather than a single service object containing all logic.

The repository has not yet committed a final implementation framework or detailed class/interface set. When those are added, the framework boundary should not silently move the rules out of Listing, StudentAccount and Order without a documented replacement decision.

11. User-interface, accessibility and usability

Final wireframes are not yet the strongest repository evidence, but UC-05 already imposes clear UI obligations: show the seller-response deadline, cancellation condition, offline payment/handover disclosure, current order state and a useful refusal message when the listing became unavailable.

- Do not hide the offline-payment boundary inside difficult-to-find terms.
- Clearly distinguish Available, Reserved, Sold/Removed and Pending Approval states.
- Provide a readable reason when reservation is refused.
- Do not depend on colour alone to communicate state or error.
- Keep controls/status labels usable on mobile-sized screens because the evidence shows students commonly trade through phones.

12. Implementation vertical slice

The approved vertical slice is the reservation/order workflow centred on UC-05. The repository already contains editable/readable model sources for system context, domain model and core sequence. The implementation slice should preserve the same names and transitions so model-to-code reconciliation remains possible.

Artefact	Repository evidence
System context	Models/system-context.svg (editable source: Models/system-context.mmd)
Domain model	Models/domain-model.svg (editable source: Models/domain-model.mmd)
Core sequence	Models/sequence-core-use-case.svg (editable source: Models/sequence-core-use-case.mmd)
Use-case model	Models/usecasemodel.svg (editable source: Models/usecasemodel.drawio)
Decision record	Decisions/D-001.md
Consistency matrix	Docs/consistency-matrix.md
Traceability matrix	Docs/traceability-matrix.md
Order state machine	Models/order-state-machine.mmd and Models/order-state-machine.svg

13. Testing and results

Phase 1 mainly defines verification intent. The most important tests are requirement-linked success, validation and failure cases for the reservation lifecycle.

Test	Scenario	Expected result
T-01 / FR-05	Eligible buyer reserves Available listing	Exactly one Pending order; listing becomes Reserved.
T-02 / FR-05	Two buyers confirm same listing nearly simultaneously	Exactly one succeeds; second receives unavailable/refusal result.
T-03 / FR-05	Listing becomes unavailable before confirmation	No new Order; listing state remains unchanged.
T-04 / FR-06	Seller rejects	Order -> Rejected; listing -> Available.
T-05 / FR-07	Seller does not respond for 24h	Order -> Expired; listing -> Available.
T-06 / FR-07	Buyer cancels before handover	Order -> Cancelled; listing -> Available.
T-07 / FR-08	Seller marks Delivered, buyer confirms	Order -> Completed.
T-08 / FR-09	Review attempted before completion	Rejected.
T-09 / FR-09	Review after completion	One review accepted; second refused.
T-10 / notification quality	Seller notice fails	Order remains Pending; failure recorded for retry.

14. Reconciliation and change history

The current repository shows active reconciliation work. The newer domain and sequence models use a direct reservation flow and separate domain rules from control logic. Some earlier artefacts still contain cart-based wording or older lifecycle terms. Those documents should be reconciled before the Phase 1 tag so the repository describes one consistent project.

Reconciled issue	Final Phase 1 resolution
Cart wording in older artefacts	Direct UC-05 reservation is the baseline; cart wording must be removed from older scope/acceptance artefacts.
Order/listing state vocabulary	Order: Pending, Accepted, Rejected, Expired, Cancelled, Delivered, Completed. Listing: Pending Approval, Available, Reserved, Sold/Removed.
Administrator modelling	PlatformAdministrator is a distinct analysis role for moderation and report resolution.
System-caused expiry history	OrderStatusChange uses actorType=System for automatic 24-hour expiry.
Listing terminal state	Sold is used after successful completion; Removed is reserved for moderation/withdrawal.

15. Risks, technical debt and release recommendation

Risk	Current control	Recommendation
Unauthorised/fake users	Synthetic verification plus account status checks	Do not imply live UB integration.
Prohibited/stolen items	Approval, low-risk scope, reporting/moderation	Keep prohibition rules explicit and testable.
Double reservation/stale page	Second availability check + one live-order constraint	Carry this into database/implementation design.
Offline transaction safety	D-001 disclosure + two-party confirmation	Recommend public campus handover locations and clarify platform limitation.
Notification failure	Record delivery outcome/retry	Never roll back valid reservation solely because notice failed.
Model inconsistency	Traceability and consistency matrices	Close naming/state differences before final Phase 1 tag.

Release recommendation: suitable for final Phase 1 review once remaining terminology conflicts are reconciled and any required lifecycle/state-machine evidence is committed. The UC-05 vertical slice is coherent, but historical cart/state wording should not remain unresolved at submission.

16. Academic integrity and AI-use record

Generative AI assistance was used to help organise this report, cross-check the public repository and re-render repository Mermaid model content into figures suitable for the Word report. The team remains responsible for verifying the final wording, requirements, decisions, model semantics and repository evidence against the lecturer's instructions. Any course-specific AI declaration should be completed by the team before submission.

References

1. University of Botswana. CSI473 Project Design Report Template, Semester 1, 2026/27.
2. TRANS-UB GitHub repository: https://github.com/lebogang00279/Trans-UB
3. TRANS-UB, Decisions/D-001.md - Offline Payment and Physical Handover.
4. TRANS-UB, Models/system-context.md and Models/system-context.mmd.
5. TRANS-UB, Models/domain-model.md and Models/domain-model.mmd.
6. TRANS-UB, Models/sequence-core-use-case.md and Models/sequence-core-use-case.mmd.
7. TRANS-UB, Evidence/problem-evidence.md and associated Evidence images.
8. TRANS-UB, Docs/requirements.md, Docs/consistency-matrix.md and Docs/traceability-matrix.md.

Appendices

Appendix A - Repository figure sources

Figure	Repository source	Use
Figure 1	Models/system-context.svg	Repository system-context export; report figure re-rendered from its source definition.
Figure 2	Models/usecasemodel.svg	Repository use-case export; report figure re-created from Models/usecasemodel.drawio.
Figure 3	Models/domain-model.svg	Repository domain-model export; report figure re-rendered from its source definition.
Figure 4	Models/sequence-core-use-case.svg	Repository sequence export; report figure re-rendered from its source definition.
Figure 5	Requirements + UC-05 + D-001	Report-only lifecycle summary derived from repository content.

Appendix B - D-001 transaction completion flow

1. Buyer places/reserves the order.
2. Seller accepts the order and arranges physical handover directly with the buyer.
3. Seller marks the order as Delivered.
4. Buyer confirms that the item was received.
5. Order becomes Completed.
6. Only then may the buyer submit a review.